

CS224W Homework 2

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1 GCN

Let $G = (V, E)$ have node features $\mathbf{x}(v)$. Initialize $\mathbf{h}_v^{(0)} = \mathbf{x}(v)$, and for each iteration k update

$$\begin{aligned}\mathbf{h}_{\mathcal{N}(v)}^{(k)} &= \text{aggregate} \left(\left\{ \mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v) \right\} \right), \\ \mathbf{h}_v^{(k)} &= \text{combine} \left(\mathbf{h}_v^{(k-1)}, \mathbf{h}_{\mathcal{N}(v)}^{(k)} \right).\end{aligned}$$

The argument of aggregate is a multiset. A graph-level embedding is obtained using

$$\text{readout} \left(\left\{ \mathbf{h}_v^{(K)} : v \in V \right\} \right).$$

1.1 Graph Isomorphism

Determine whether the two graphs shown in Question 1.1 of the homework PDF are isomorphic. If they are, give a bijection between their vertices that preserves edges. If they are not, identify a structure present in only one graph.

Solution. The two graphs are isomorphic. Define the bijection

v	1	2	3	4	5	6	7	8
$\phi(v)$	A	D	E	H	C	B	G	F

from the numbered graph to the lettered graph. It is enough to check the neighborhoods of nodes 1, 2, 3, 4, since every edge in the numbered graph has one endpoint in this set:

$$\begin{aligned}\phi(\mathcal{N}(1)) &= \phi(\{5, 6, 7\}) = \{C, B, G\} = \mathcal{N}(A), \\ \phi(\mathcal{N}(2)) &= \phi(\{5, 6, 8\}) = \{C, B, F\} = \mathcal{N}(D), \\ \phi(\mathcal{N}(3)) &= \phi(\{5, 7, 8\}) = \{C, G, F\} = \mathcal{N}(E), \\ \phi(\mathcal{N}(4)) &= \phi(\{6, 7, 8\}) = \{B, G, F\} = \mathcal{N}(H).\end{aligned}$$

Thus, for every pair u, v ,

$$\{u, v\} \in E_1 \iff \{\phi(u), \phi(v)\} \in E_2.$$

Therefore ϕ is an isomorphism.

1.2 Aggregation Expressiveness

Consider the three aggregation functions

$$\begin{aligned} \left[\text{aggregate}_{\max} \left(\left\{ \mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v) \right\} \right) \right]_i &= \max_{u \in \mathcal{N}(v)} \left[\mathbf{h}_u^{(k-1)} \right]_i, \\ \text{aggregate}_{\text{mean}} \left(\left\{ \mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v) \right\} \right) &= \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(k-1)}, \\ \text{aggregate}_{\text{sum}} \left(\left\{ \mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v) \right\} \right) &= \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(k-1)}. \end{aligned}$$

Give two graphs G_1, G_2 , initial node features, and nodes $v_1 \in V_1, v_2 \in V_2$ such that $\mathbf{h}_{v_1}^{(0)} = \mathbf{h}_{v_2}^{(0)}$. After one update, the two nodes must remain equal under mean and max aggregation but become different under sum aggregation.

Solution. Let G_1 contain one edge $\{v_1, u\}$, and let G_2 contain the two edges $\{v_2, a\}$ and $\{v_2, b\}$. Assign scalar initial features

$$h_{v_1}^{(0)} = h_{v_2}^{(0)} = 0, \quad h_u^{(0)} = h_a^{(0)} = h_b^{(0)} = 1,$$

and choose

$$\text{combine}(x, y) = x + y.$$

The neighbor-feature multisets of the distinguished nodes are

$$\left\{ h_u^{(0)} : u \in \mathcal{N}(v_1) \right\} = \{1\}, \quad \left\{ h_u^{(0)} : u \in \mathcal{N}(v_2) \right\} = \{1, 1\}.$$

Consequently,

aggregator	$h_{\mathcal{N}(v_1)}^{(1)}$	$h_{\mathcal{N}(v_2)}^{(1)}$
max	1	1
mean	1	1
sum	1	2

Since both distinguished nodes have initial feature zero, the combine step does not change these values. Hence max and mean give equal updated features, whereas sum gives

$$h_{v_1}^{(1)} = 1 \neq 2 = h_{v_2}^{(1)}.$$

The example works because max and mean discard the multiplicity of repeated neighbor features, while sum preserves it.

1.3 Relation to the Weisfeiler–Lehman Test

At each iteration, the Weisfeiler–Lehman (WL) test updates each node representation using its previous representation together with the multiset of its neighbors' previous representations.

Prove that the neural graph-isomorphism test above is at most as powerful as the WL test. More precisely, for two non-isomorphic graphs updated for K iterations using the same aggregate and combine, show that

$$\text{readout} \left(\left\{ \mathbf{h}_v^{(K)} : v \in V_1 \right\} \right) \neq \text{readout} \left(\left\{ \mathbf{h}_v^{(K)} : v \in V_2 \right\} \right)$$

implies that the WL test also declares the graphs non-isomorphic.

Solution. Let $l_v^{(k)}$ be the WL label of node v after iteration k . The WL hash is injective on the pair consisting of the node's previous label and the multiset of its neighbors' previous labels. We first prove by induction that, for nodes from either graph,

$$l_u^{(k)} = l_v^{(k)} \implies \mathbf{h}_u^{(k)} = \mathbf{h}_v^{(k)}. \quad (1)$$

For $k = 0$, both algorithms initialize from the same node feature:

$$l_v^{(0)} = \mathbf{x}(v) = \mathbf{h}_v^{(0)}.$$

Thus (1) holds at iteration zero.

Assume (1) holds after iteration $k - 1$, and suppose $l_u^{(k)} = l_v^{(k)}$. Injectivity of the WL hash implies both

$$l_u^{(k-1)} = l_v^{(k-1)}$$

and equality of multisets

$$\{l_a^{(k-1)} : a \in \mathcal{N}(u)\} = \{l_b^{(k-1)} : b \in \mathcal{N}(v)\}.$$

By the induction hypothesis, equal previous WL labels have equal previous neural embeddings. Applying this implication element-by-element, with multiplicity, gives

$$\{\mathbf{h}_a^{(k-1)} : a \in \mathcal{N}(u)\} = \{\mathbf{h}_b^{(k-1)} : b \in \mathcal{N}(v)\}.$$

The shared, permutation-invariant aggregate function therefore produces the same neighborhood embedding for u and v . Their own previous embeddings are also equal by the induction hypothesis, so the shared combine function gives

$$\mathbf{h}_u^{(k)} = \mathbf{h}_v^{(k)}.$$

This completes the induction.

Now argue by contradiction. Suppose the WL test does not distinguish G_1 and G_2 after K iterations. Then their final WL-label multisets, including multiplicities, are equal:

$$\{l_v^{(K)} : v \in V_1\} = \{l_v^{(K)} : v \in V_2\}.$$

By (1), a node's neural embedding is a deterministic function of its WL label. Hence the two final neural-embedding multisets are also equal:

$$\{\mathbf{h}_v^{(K)} : v \in V_1\} = \{\mathbf{h}_v^{(K)} : v \in V_2\}.$$

Applying the same readout function to equal multisets must give equal graph embeddings, contradicting the hypothesis that the two readouts differ. Therefore, whenever the neural model distinguishes the graphs, the WL test also distinguishes them. The neural model is thus at most as powerful as the WL test.

2 Node Embeddings with TransE

A multi-relational graph is $G = (\mathcal{E}, \mathcal{S}, \mathcal{L})$, where $\mathcal{E}$ is the entity set, $\mathcal{L}$ is the relation set, and $\mathcal{S}$ contains triples (h, ℓ, t) . TransE learns entity and relation embeddings in $\mathbb{R}^k$ so that

$$\mathbf{h} + \boldsymbol{\ell} \approx \mathbf{t}$$

for observed triples.

For a positive triple, define its corruption set by

$$\mathcal{S}'_{(h, \ell, t)} = \{(h', \ell, t) : h' \in \mathcal{E}\} \cup \{(h, \ell, t') : t' \in \mathcal{E}\}.$$

The TransE margin loss is

$$\mathcal{L} = \sum_{(h, \ell, t) \in \mathcal{S}} \sum_{(h', \ell, t') \in \mathcal{S}'_{(h, \ell, t)}} [\gamma + d(\mathbf{h} + \boldsymbol{\ell}, \mathbf{t}) - d(\mathbf{h}' + \boldsymbol{\ell}, \mathbf{t}')]_+,$$

where d is Euclidean distance, $[a]_+ = \max(0, a)$, and entity embeddings are constrained by $\|\mathbf{e}\|_2 = 1$.

2.1 Simplified Loss

Suppose the objective is replaced by

$$\mathcal{L}_{\text{simple}} = \sum_{(h, \ell, t) \in \mathcal{S}} d(\mathbf{h} + \boldsymbol{\ell}, \mathbf{t}).$$

Give a non-trivial graph with at least two nodes and one edge, together with two-dimensional embeddings, that achieves zero loss but is completely uninformative.

Solution. Let

$$\mathcal{E} = \{a, b\}, \quad \mathcal{L} = \{r\}, \quad \mathcal{S} = \{(a, r, b)\}.$$

Choose the two-dimensional embeddings

$$\mathbf{a} = \mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The entity embeddings satisfy the unit-norm constraint. Moreover,

$$d(\mathbf{a} + \mathbf{r}, \mathbf{b}) = \left\| \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\|_2 = 0,$$

so $\mathcal{L}_{\text{simple}} = 0$.

This representation is useless because the two distinct entities have exactly the same embedding. In fact, for every $x, y \in \{a, b\}$,

$$\mathbf{x} + \mathbf{r} = \mathbf{y},$$

so the embeddings predict every possible r -triple and cannot determine which edge is actually present.

2.2 Removing the Margin

Now remove γ and optimize

$$\mathcal{L}_{\text{no margin}} = \sum_{(h,\ell,t) \in \mathcal{S}} \sum_{(h',\ell,t') \in \mathcal{S}'_{(h,\ell,t)}} [d(\mathbf{h} + \boldsymbol{\ell}, \mathbf{t}) - d(\mathbf{h}' + \boldsymbol{\ell}, \mathbf{t}')]_+.$$

Give a non-trivial graph and two-dimensional embeddings that achieve zero loss but do not reveal whether a pair of entities is connected by a relation.

Solution. Use the same graph

$$\mathcal{E} = \{a, b\}, \quad \mathcal{L} = \{r\}, \quad \mathcal{S} = \{(a, r, b)\},$$

but choose

$$\mathbf{a} = \mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{r} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The relation embedding is now nonzero. The positive-triple distance is

$$d(\mathbf{a} + \mathbf{r}, \mathbf{b}) = 1.$$

Every corrupted head or tail is either a or b , and both have the same embedding. Hence every corrupted triple also has distance

$$d(\mathbf{h}' + \mathbf{r}, \mathbf{t}') = 1.$$

Each summand in the no-margin loss is therefore

$$[1 - 1]_+ = 0,$$

so $\mathcal{L}_{\text{no margin}} = 0$.

Nevertheless, all entity pairs receive the same score. The objective only requires a positive triple to be no farther away than a corrupted triple; in the absence of γ , equality is sufficient. The positive margin is what rules out this all-scores-tied solution.

2.3 Removing Entity Normalization

TransE normally renormalizes every entity embedding to unit length. Describe how the algorithm could trivially reduce the loss if this normalization step were omitted, and characterize the resulting embeddings.

Solution. Consider any collection of unnormalized embeddings $\{\bar{\mathbf{e}}\}$ and $\{\bar{\boldsymbol{\ell}}\}$. Scale every entity and relation by a common factor $c > 0$:

$$\mathbf{e} = c\bar{\mathbf{e}}, \quad \boldsymbol{\ell} = c\bar{\boldsymbol{\ell}}.$$

For one positive triple and one of its corruptions, let

$$d_+ = d(\bar{\mathbf{h}} + \bar{\boldsymbol{\ell}}, \bar{\mathbf{t}}), \quad d_- = d(\bar{\mathbf{h}}' + \bar{\boldsymbol{\ell}}, \bar{\mathbf{t}}').$$

Euclidean distance is homogeneous, so the corresponding loss term after scaling is

$$[\gamma + cd_+ - cd_-]_+ = [\gamma - c(d_- - d_+)]_+.$$

Whenever $d_- > d_+$, choosing

$$c \geq \frac{\gamma}{d_- - d_+}$$

makes this term zero. For a finite training set whose positive-negative gaps are all strictly positive, one sufficiently large c makes every hinge term zero.

Thus the optimizer can satisfy the fixed margin partly by increasing the scale of all embeddings instead of learning a better normalized geometry. The entity and relation norms can grow without bound, points become arbitrarily far from the origin, and absolute distances become meaningless and numerically unstable. Unit normalization removes this scale degree of freedom and forces the model to improve relative directions and positions within a bounded space.

2.4 Limits of TransE Expressiveness

Give a simple multi-relational graph for which no perfect embedding exists: there is no assignment satisfying

$$\mathbf{u} + \ell = \mathbf{v} \quad \text{for all } (u, \ell, v) \in \mathcal{S}$$

while also satisfying

$$\mathbf{u} + \ell \neq \mathbf{v} \quad \text{for all } (u, \ell, v) \notin \mathcal{S}.$$

Use two-dimensional embeddings and explain what the example shows about the types of relations TransE can and cannot model.

Solution. Let

$$\mathcal{E} = \{a, b, c\}, \quad \mathcal{L} = \{r, s\},$$

and let the only positive triples be

$$\mathcal{S} = \{(a, r, b), (a, r, c), (b, s, a)\}.$$

In particular, $(c, s, a) \notin \mathcal{S}$.

If a perfect TransE embedding existed, the first two positive triples would require

$$\mathbf{a} + \mathbf{r} = \mathbf{b}, \quad \mathbf{a} + \mathbf{r} = \mathbf{c}.$$

Therefore

$$\mathbf{b} = \mathbf{c}.$$

The third positive triple requires

$$\mathbf{b} + \mathbf{s} = \mathbf{a}.$$

Substituting $\mathbf{c} = \mathbf{b}$ then gives

$$\mathbf{c} + \mathbf{s} = \mathbf{a}.$$

This predicts (c, s, a) as a positive triple, contradicting $(c, s, a) \notin \mathcal{S}$. Hence no perfect embedding exists in $\mathbb{R}^2$; in fact, the contradiction holds in every dimension.

The failure comes from the fixed-translation form of TransE. For a given head embedding and relation embedding, $\mathbf{h} + \mathbf{r}$ is a single point. A one-to-many relation such as

$$(a, r, b), \quad (a, r, c)$$

therefore forces all of its tails to share one embedding. If those tails must play different roles elsewhere in the graph, TransE necessarily creates false triples. Thus TransE naturally represents one-to-one translations but cannot faithfully express general one-to-many, many-to-one, or many-to-many relations while keeping the involved entities distinguishable.

3 Expressive Power of Knowledge Graph Embeddings

Consider the following common relational patterns:

- **Symmetry:** A is married to B , and B is married to A .
- **Inverse:** A is teacher of B , and B is student of A .
- **Composition:** A is son of B , and C is sister of B , so C is aunt of A .

3.1 Relational Patterns in TransE

For each pattern above, determine whether TransE can model it perfectly with

$$\mathbf{h} + \mathbf{\ell} = \mathbf{t}$$

for every relation. Explain each answer. A zero relation embedding is not allowed because it makes related entities indistinguishable.

Solution. Write $\mathbf{m}$, $\mathbf{t}$, and $\mathbf{s}$ for the embeddings of *married to*, *teacher of*, and *student of*, respectively.

Symmetry cannot be modeled. For two distinct entities A and B , symmetry would require

$$\mathbf{a} + \mathbf{m} = \mathbf{b}, \quad \mathbf{b} + \mathbf{m} = \mathbf{a}.$$

Substituting the first equation into the second gives

$$\mathbf{a} + 2\mathbf{m} = \mathbf{a},$$

so $\mathbf{m} = \mathbf{0}$. The first equation then also gives $\mathbf{a} = \mathbf{b}$. This is precisely the forbidden degenerate solution; therefore TransE cannot represent a non-trivial symmetric relation perfectly.

Inverse relations can be modeled. The two facts require

$$\mathbf{a} + \mathbf{t} = \mathbf{b}, \quad \mathbf{b} + \mathbf{s} = \mathbf{a}.$$

They are consistent exactly when

$$\mathbf{s} = -\mathbf{t}.$$

For example, take

$$\mathbf{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \mathbf{s} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Both entities have unit norm, both relations are nonzero, and the two equations hold.

Composition can be modeled. Let $\mathbf{r}_{\text{son}}$, $\mathbf{r}_{\text{sister}}$, and $\mathbf{r}_{\text{aunt}}$ denote the three relation embeddings. The facts in the question require

$$\mathbf{a} + \mathbf{r}_{\text{son}} = \mathbf{b}, \quad \mathbf{c} + \mathbf{r}_{\text{sister}} = \mathbf{b}, \quad \mathbf{c} + \mathbf{r}_{\text{aunt}} = \mathbf{a}.$$

Consequently, the required composition rule is

$$\mathbf{r}_{\text{sister}} = \mathbf{r}_{\text{aunt}} + \mathbf{r}_{\text{son}}.$$

For a concrete unit-norm realization, choose

$$\mathbf{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -1 \\ 0 \end{bmatrix},$$

and set

$$\mathbf{r}_{\text{son}} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \mathbf{r}_{\text{sister}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{r}_{\text{aunt}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

All three relations are nonzero and all three facts are satisfied exactly.

3.2 Relational Patterns in RotatE

RotatE represents each entity as d two-dimensional points, equivalently a vector in $\mathbb{R}^{2d}$, and represents a relation by d rotation angles. For each $i = 0, \dots, d-1$, applying relation ℓ rotates (h_{2i}, h_{2i+1}) clockwise by angle ℓ_i . Entity embeddings have unit ℓ_2 norm.

Determine whether RotatE can model symmetry, inverse relations, and composition perfectly, and justify each answer.

Solution. Let R_θ denote clockwise rotation by angle θ . Rotations compose by adding their angles:

$$R_\alpha R_\beta = R_{\alpha+\beta},$$

where angles are understood modulo 2π , coordinatewise when $d > 1$.

Symmetry can be modeled. It requires

$$R_m \mathbf{a} = \mathbf{b}, \quad R_m \mathbf{b} = \mathbf{a}.$$

Thus applying the relation twice must give the identity. A nonzero choice is $m = \pi$. For example, with $d = 1$,

$$\mathbf{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ 0 \end{bmatrix},$$

a rotation by π swaps A and B .

Inverse relations can be modeled. If the teacher relation has angle t , choose the student relation to have angle

$$s \equiv -t \pmod{2\pi}.$$

Then $R_s R_t = I$. For instance, $t = \pi/2$ and $s = -\pi/2$ are both nonzero and map an entity to another unit vector and back.

Composition can be modeled. The three facts give

$$R_{\mathbf{r}_{\text{son}}} \mathbf{a} = \mathbf{b}, \quad R_{\mathbf{r}_{\text{sister}}} \mathbf{c} = \mathbf{b}, \quad R_{\mathbf{r}_{\text{aunt}}} \mathbf{c} = \mathbf{a}.$$

Therefore it suffices to choose

$$\mathbf{r}_{\text{sister}} \equiv \mathbf{r}_{\text{aunt}} + \mathbf{r}_{\text{son}} \pmod{2\pi}.$$

For example, take $d = 1$, $\mathbf{c} = (1, 0)^\top$, $\mathbf{r}_{\text{aunt}} = \pi/3$, $\mathbf{r}_{\text{son}} = \pi/2$, and $\mathbf{r}_{\text{sister}} = 5\pi/6$. Define $\mathbf{a} = R_{\pi/3} \mathbf{c}$ and $\mathbf{b} = R_{\pi/2} \mathbf{a}$. All entities have unit norm, all relation angles are nonzero, and the composition holds exactly.

3.3 A Limitation of RotatE

Give an example of a graph that RotatE cannot model perfectly. Determine whether TransE can model the same graph. Assume relation embeddings cannot be zero in either model.

Solution. Let

$$\mathcal{E} = \{a, b, c, d\}, \quad \mathcal{L} = \{r, s\},$$

and let the only positive triples be

$$\mathcal{S} = \{(a, r, b), (a, r, c), (b, s, d)\}.$$

In particular, $(c, s, d) \notin \mathcal{S}$.

Suppose RotatE had a perfect embedding, and write R_r and R_s for the blockwise rotations associated with the two relations. The first two positive triples imply

$$R_r \mathbf{a} = \mathbf{b}, \quad R_r \mathbf{a} = \mathbf{c},$$

and hence $\mathbf{b} = \mathbf{c}$. The third positive triple then implies

$$R_s \mathbf{c} = R_s \mathbf{b} = \mathbf{d}.$$

Thus RotatE necessarily predicts the nonexistent triple (c, s, d) , a contradiction. This proof holds in every dimension and does not rely on either relation being zero.

TransE cannot model this graph perfectly either. Indeed,

$$\mathbf{a} + \mathbf{r} = \mathbf{b}, \quad \mathbf{a} + \mathbf{r} = \mathbf{c}$$

again forces $\mathbf{b} = \mathbf{c}$. From $\mathbf{b} + \mathbf{s} = \mathbf{d}$, it follows that

$$\mathbf{c} + \mathbf{s} = \mathbf{d},$$

which incorrectly predicts (c, s, d) . Both models apply a fixed, deterministic transformation to a head entity for each relation, so neither can perfectly represent this one-to-many relation while keeping B and C distinguishable.

Honor Code

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